THE COTANGENT COMPLEX OF A LOCAL COMPLETE INTERSECTION MAP IS PERFECT WITH TOR AMPLITUDE IN [-1, 0]

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cotangent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement is two sentences; the proof separates the perfectness statement from the concentration of cohomology and closes with the distinguished triangle.

Editorial note. The five references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the retained preamble of this chapter (sources/stacks-cotangent/retained-preamble.tex, lines 1-63), containing the Xy-pic setup, the comment environments, the theorem declarations and the macros including \NL.

Allowed standard prerequisites (editorial).

- The identification of the naive cotangent complex with the truncated cotangent complex. Lemma lemma-relation-with-naive-cotangent-complex, used at proof lines 2965-2968. Turns the two-term presentation complex into the truncation.
- Finite projectivity of the conormal module of a quasi-regular ideal. More on Algebra, Lemma more-algebra-lemma-quasi-regular-ideal-finite-projective, used at proof lines 2969-2973. Makes the truncation a perfect complex with tor amplitude in [-1, 0].
- The distinguished triangle for the composition A -> P -> B. equation-triangle, used at proof lines 2975-2981. Reduces the vanishing of the remaining cohomology groups to that of L_{B/P}.
- The Koszul regularity of the ideal in the local complete intersection case and the polynomial algebra computation. Lemmas lemma-mod-Koszul-regular-ideal and lemma-cotangent-complex-polynomial-algebra, used at proof lines 2980-2983. Give the concentration in degrees -1 and 0 and the computation of the left term.

Source reference. Original equation 08QS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11070–11070. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08QH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11060–11060. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:cotangent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 08SK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11133–11133. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08RB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11089–11089. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08RK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11098–11098. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Proposition 0.2. *Let $A \to B$ be a local complete intersection map. Then $L_{B/A}$ is a perfect complex with tor amplitude in $[-1, 0]$.*

Proof. Choose a surjection $P = A[x_1, \ldots, x_n] \to B$ with kernel J. By Lemma 08RB we see that $J/J^2 \to \bigoplus B\mathrm{d}x_i$ is quasi-isomorphic to $\tau_{\geq -1} L_{B/A}$. Note that J/J^2 is finite projective (More on Algebra, Lemma 08RK), hence $\tau_{\geq -1} L_{B/A}$ is a perfect complex with tor amplitude in $[-1, 0]$. Thus it suffices to show that $H^i(L_{B/A}) = 0$ for $i \notin [-1, 0]$. This follows from (08QS)

$$L_{P/A} \otimes^{\mathbf{L}}_P B \to L_{B/A} \to L_{B/P} \to L_{P/A} \otimes^{\mathbf{L}}_P B[1]$$

and Lemma 08SK to see that $H^i(L_{B/P})$ is zero unless $i \in \{-1, 0\}$. (We also use Lemma 08QH for the term on the left.) □